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### Display apparatus having pixel defining layer covering pixel electrode edge portions

#### Abstract

A method of manufacturing a light-emitting display apparatus and a light-emitting display apparatus are provided. The method includes forming a first photosensitive layer on a conductive material layer, forming a pixel electrode by etching the conductive material layer by using the first photosensitive layer as a mask, ashing the first photosensitive layer disposed on the pixel electrode, forming a pixel defining layer that covers an edge portion of the pixel electrode and includes a first opening overlapping the ashed first photosensitive layer, removing the ashed first photosensitive layer disposed in the first opening, forming an intermediate layer including a functional layer and an emission layer on the pixel defining layer, and forming an opposite electrode on the intermediate layer.

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#### Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This is a divisional application of U.S. patent application Ser. No. 16/899,124, filed Jun. 11, 2020, issued on Nov. 15, 2022, as U.S. Pat. No. 11,502,145, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 16/899,124 claims priority to and benefits of Korean Patent Application

## BACKGROUND

### 1. Technical Field

(1) One or more embodiments relate to a display apparatus, and more particularly, to a light-emitting display apparatus.

### 2. Description of the Related Art

(2) An organic light-emitting diode display apparatus, which is a display apparatus for displaying images, has self-emission characteristics unlike a liquid crystal display apparatus. Therefore, since the organic light-emitting diode display apparatus does not require a separate light source, the thickness and weight of the organic light-emitting diode display apparatus may be reduced. Also, the organic light-emitting diode display apparatus may have high quality characteristics such as low power consumption, high luminance, and high response time.

## SUMMARY

(3) An organic light-emitting diode display apparatus as described above should have excellent contrast and luminance, but the contrast may be poor in bright external light. In order to prevent this limitation, a pixel defining layer having a certain color (for example, black or the like) may be formed. Dark spots may be generated by residues that occur in a process of forming a colored pixel defining layer. One or more embodiments may provide a light-emitting display apparatus and a manufacturing method thereof, whereby deterioration of characteristics for external light and defects caused by generation of dark spots may be prevented and manufacturing processes may be simplified. However, these limitations are merely examples, and the scope of the disclosure is not limited thereto.

(4) Additional aspects will be set forth in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the embodiments of the invention.

(5) According to embodiments, a method of manufacturing a light-emitting display apparatus includes forming a first photosensitive layer on a conductive material layer, forming a pixel electrode by etching the conductive material layer by using the first photosensitive layer as a mask, ashing the first photosensitive layer disposed on the pixel electrode, forming a pixel defining layer that covers an edge portion of the pixel electrode and including a first opening overlapping the ashed first photosensitive layer, removing the ashed first photosensitive layer disposed in the first opening, forming an intermediate layer including a functional layer and an emission layer on the pixel defining layer, and forming an opposite electrode on the intermediate layer.

(6) In an embodiment, the forming of the pixel defining layer may include forming a color material layer on the ashed first photosensitive layer, exposing a part of the color material layer, and forming the first opening by developing the exposed part of the color material layer.

(7) The pixel defining layer may include a negative photosensitive material.

(8) The method may further include curing the pixel defining layer after the forming of the first opening.

(9) The pixel defining layer may further include a color pigment or carbon black.

(10) In an embodiment, a shape of the first opening may be substantially the same as a shape of the pixel electrode in a plan view.

(11) In an embodiment, the method may further include forming a thin-film transistor, and forming an insulating layer between the thin-film transistor and the pixel electrode. The forming of the insulating layer may include forming a contact hole for electrical connection between the thin-film transistor and the pixel electrode, and the first opening may overlap the contact hole.

(12) The method may further include forming an optical functional layer including a color filter overlapping the first opening and a black matrix adjacent to the color filter, wherein the black matrix may overlap the contact hole.

- (13) In an embodiment, the method may further include forming a second photosensitive layer on the conductive material layer, and forming a wiring by etching the conductive material layer by using the second photosensitive layer as a mask.
- (14) The method may further include ashing the second photosensitive layer disposed on the wiring, wherein the forming of the pixel defining layer may include forming a second opening overlapping the ashed second photosensitive layer and spaced apart from the first opening.
- (15) The method may further include removing the ashed second photosensitive layer arranged in the second opening.
- (16) The method may further include forming an insulating layer overlapping the second opening and disposed between the wiring and the intermediate layer.
- (17) The method may further include forming a hole in a portion of the functional layer, which may overlap the wiring, wherein the opposite electrode may be in electrical contact with the wiring through the hole.
- (18) According to embodiments, a light-emitting display apparatus may include a substrate, a thin-film transistor on the substrate, a pixel electrode electrically connected to the thin-film transistor, a first insulating layer disposed between the thin-film transistor and the pixel electrode, a pixel defining layer covering an edge portion of the pixel electrode and including a first opening overlapping the pixel electrode, an intermediate layer including an emission layer overlapping the first opening and a functional layer, and an opposite electrode on the intermediate layer.
- (19) In an embodiment, the first insulating layer may include a contact hole for electrical connection between the thin-film transistor and the pixel electrode, and the first opening may overlap the contact hole.
- (20) The light-emitting display apparatus may further include an optical functional layer including a color filter overlapping the first opening, and a black matrix adjacent to the color filter and overlapping a part of the first opening.
- (21) The black matrix may overlap the contact hole.
- (22) In an embodiment, the pixel defining layer may further include a color pigment or carbon black.
- (23) In an embodiment, a shape of the first opening may be substantially the same as a shape of the pixel electrode when seen in a direction perpendicular to the substrate.
- (24) In an embodiment, a first distance from a first edge of the pixel electrode to a first edge of the first opening may be substantially the same as a second distance from a second edge of the pixel electrode to a second edge of the first opening.
- (25) In an embodiment, the light-emitting display apparatus may further include a wiring spaced apart from the pixel electrode, wherein the wiring and the pixel electrode may include a same material.
- (26) The pixel defining layer may further include a second opening covering an edge portion of the wiring and overlapping the wiring.
- (27) A shape of the second opening may be substantially the same as a shape of the wiring when seen in a direction perpendicular to the substrate.
- (28) The light-emitting display apparatus may further include a second insulating layer overlapping the second opening.
- (29) The second insulating layer may be disposed between the wiring and the intermediate layer.
- (30) The light-emitting display apparatus may further include a spacer disposed on the pixel defining layer, wherein the second insulating layer and the spacer may include a same material.
- (31) The pixel defining layer and the spacer may include different materials.
- (32) In an embodiment, light-emitting display apparatus may further include a driving thin-film transistor electrically connected to the thin-film transistor, an initialization thin-film transistor electrically connected to a gate electrode of the driving thin-film transistor or the pixel electrode, and an initialization voltage line electrically connected to the initialization thin-film transistor,

wherein the wiring may include the initialization voltage line.

(33) In an embodiment, the functional layer may include a hole overlapping the second opening.

(34) The opposite electrode may be in electrical contact with the wiring through the hole.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIGS. 1 to 6 are schematic cross-sectional views illustrating a process of manufacturing a light-emitting display apparatus, according to an embodiment;

(3) FIGS. 7A to 7C are schematic cross-sectional views illustrating a part of the process of manufacturing the light-emitting display apparatus, according to an embodiment;

(4) FIG. 8 is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment, which shows a pixel electrode and a wiring;

(5) FIG. 9 is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment, which shows a pixel defining layer formed on the pixel electrode and the wiring;

(6) FIGS. 10A to 10C are schematic cross-sectional views illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment;

(7) FIG. 11 is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment;

(8) FIG. 12 is a schematic diagram of an equivalent circuit of a pixel illustrating a pixel circuit connected to a light-emitting diode of the light-emitting display apparatus, for example, an organic light-emitting diode, according to an embodiment;

(9) FIGS. 13A to 13E are schematic cross-sectional views illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment; and

(10) FIG. 14 is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment.

### DETAILED DESCRIPTION OF THE EMBODIMENTS

(11) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the description.

(12) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. The term “and/or” is intended to include any combination of the terms “and” and “or” for the purpose of its meaning and interpretation. For example, “A and/or B” may be understood to mean “A, B, or A and B.” The terms “and” and “or” may be used in the conjunctive or disjunctive sense and may be understood to be equivalent to “and/or”.

(13) Hereinafter, embodiments will be described in detail with reference to the accompanying drawings. In the following description, like reference numerals are assigned to like components, and redundant descriptions thereof will be omitted.

(14) It will be understood that although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

(15) As used herein, the singular forms “a,” “an,” “one,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

- (16) It will be further understood that the terms “includes,” “including,” “contains,” “containing,” “comprises,” and/or “comprising,” used herein specify the presence of stated features or elements, but do not preclude the presence or addition of one or more other features or elements.
- (17) It will be understood that when a layer, region, or element is referred to as being “formed on,” another layer, region, or element, it can be directly or indirectly formed on the other layer, region, or element. For example, intervening layers, regions, or elements may be present.
- (18) Sizes of elements in the drawings may be exaggerated for convenience of explanation. For example, since sizes and thicknesses of elements in the drawings are arbitrarily illustrated for convenience of explanation, the following embodiments are not limited thereto.
- (19) When a certain embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order.
- (20) In the following embodiments, it will be understood that when a film, layer, region, element, or component is referred to as being “connected to” or “coupled to” another film, layer, region, element, and component, it may be directly or indirectly connected or coupled to the other film, layer, region, element, or component. For example, intervening films, regions, or components may be present. In the following embodiments, it will be understood that when a film, layer, region, element, or component is referred to as being “electrically connected to” or “electrically coupled to” another film, layer, region, element, and component, it may be directly or indirectly electrically connected or coupled to the other film, layer, region, element, or component. For example, intervening films, layers, regions, elements, or components may be present.
- (21) FIGS. **1** to **6** are schematic cross-sectional views illustrating a process of manufacturing a light-emitting display apparatus, according to an embodiment.
- (22) Referring to FIG. **1**, a conductive material layer **200** is formed above a substrate **100**, and a first photosensitive layer **1100** and a second photosensitive layer **1200** are formed on the conductive material layer **200**. In one embodiment, FIG. **1** illustrates that the conductive material layer **200** is formed on a planarization insulating layer **170**. Before the conductive material layer **200** is formed, a thin-film transistor TFT and a storage capacitor Cap may be formed on the substrate **100**.
- (23) The substrate **100** may include various materials, such as glass material, a metal material, or a plastic material (for example, polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polyimide, or the like). A buffer layer **110**, a gate insulating layer **130**, an interlayer insulating layer **150**, and the planarization insulating layer **170** may be formed on the substrate **100**. The buffer layer **110** prevents impurities from penetrating into a semiconductor layer of the thin-film transistor TFT. The gate insulating layer **130** insulates the semiconductor layer and a gate electrode of the thin-film transistor TFT from each other. The interlayer insulating layer **150** insulates a source electrode, a drain electrode, and the gate electrode of the thin-film transistor TFT from each other. The planarization insulating layer **170** covers the thin-film transistor TFT and has an approximately flat upper surface.
- (24) The conductive material layer **200** may include a reflective film including silver (Ag), magnesium (Mg), aluminum (Al), platinum (Pt), palladium (Pd), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), and/or any compound thereof. The conductive material layer **200** may include a reflective film including the above-mentioned material, and a transparent conductive film disposed above and/or below the reflective film. The transparent conductive film may include indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnO), indium oxide (In.sub.2O.sub.3), indium gallium oxide (IGO), aluminum zinc oxide (AZO), or the like. In one embodiment, the conductive material layer **200** may have a multi-layered structure in which an ITO film, an Ag film, and an ITO film are sequentially stacked.
- (25) The first photosensitive layer **1100** and the second photosensitive layer **1200** are spaced apart

from each other. The first photosensitive layer **1100** may overlap a contact hole CNT configured to electrically connect the conductive material layer **200** to the thin-film transistor TFT.

(26) The first photosensitive layer **1100** and the second photosensitive layer **1200** may include a positive photosensitive material. The first photosensitive layer **1100** and the second photosensitive layer **1200** may be formed by coating a photosensitive material layer on the conductive material layer **200** and performing an exposure process and a developing process using a mask including a light transmitting portion and a light blocking portion. A soft bake process may be performed at a first temperature between the process of coating the photosensitive material layer and the exposure process, and a post bake process may be performed at a second temperature after the first photosensitive layer **1100** and the second photosensitive layer **1200** are formed. The first temperature may be in a range of about 105° C. to about 110° C., and the second temperature may be greater than the first temperature. For example, the second temperature may be in a range of about 130° C. to about 135° C. Adhesion with the conductive material layer **200** may be increased through the post bake process.

(27) As illustrated in FIG. 2, the conductive material layer **200** may be etched to form a pixel electrode **210** and a wiring **220**. An etching process (for example, a wet etching process) may be performed on the conductive material layer **200** by using the first photosensitive layer **1100** and the second photosensitive layer **1200** as a mask. The pixel electrode **210** may be formed under the first photosensitive layer **1100** through the etching process, and the wiring **220** may be formed under the second photosensitive layer **1200** through the etching process. The pixel electrode **210** may have a planar shape corresponding to the first photosensitive layer **1100**, and the wiring **220** may have a planar shape corresponding to the second photosensitive layer **1200**. For example, the pixel electrode **210** and the wiring **220** may include a same material. The pixel electrode **210** may be electrically connected to the thin-film transistor TFT through the contact hole CNT of the planarization insulating layer **170**.

(28) As illustrated in FIG. 3, the first photosensitive layer **1100** and the second photosensitive layer **1200** on the pixel electrode **210** and the wiring **220**, respectively, may be ashed. The ashing may use oxygen. The ashed first photosensitive layer **1100** and the ashed second photosensitive layer **1200** may be reduced in size. An edge portion of the pixel electrode **210** and an edge portion of the wiring **220** may be exposed through the ashing process. The ashing is distinguished from a strip process. The ashing process may reduce the thicknesses and widths of the first photosensitive layer **1100** and the second photosensitive layer **1200** as a whole.

(29) As illustrated in FIG. 4, a color material layer **180M** may be coated on the ashed first photosensitive layer **1100** and the ashed second photosensitive layer **1200**, and the color material layer **180M** may be exposed by using a mask **1400**. The mask **1400** may include a light transmitting portion **1401** and a light blocking portion **1402**.

(30) The color material layer **180M** may be coated on the entire substrate **100** to cover the first photosensitive layer **1100** and the second photosensitive layer **1200**. The color material layer **180M** may have different photosensitivity from the first photosensitive layer **1100** and the second photosensitive layer **1200**. The color material layer **180M** may include a negative photosensitive material.

(31) The light transmitting portion **1401** of the mask **1400** may not overlap the first photosensitive layer **1100** and the second photosensitive layer **1200**, and the light blocking portion **1402** may overlap the first photosensitive layer **1100** and the second photosensitive layer **1200**. Therefore, the portion of the color material layer **180M** overlapping the first photosensitive layer **1100** and the second photosensitive layer **1200** may not be exposed, and the remaining portion of the color material layer **180M** may be exposed.

(32) Among the light blocking portions **1402** of the mask **1400**, a first width **W1** of the light blocking portion **1402** overlapping the first photosensitive layer **1100** may be equal to or similar to a second width **W2** of the first photosensitive layer **1100**. That the first width **W1** is similar to the



second width **W2** may mean that a difference between the first width **W1** and the second width **W2** is less than about 10% of the first width **W1** or less than about 10% of the second width **W2**.

(33) Among the light blocking portions **1402** of the mask **1400**, a third width **W3** of the light blocking portion **1402** overlapping the second photosensitive layer **1200** may be equal to or similar to a fourth width **W4** of the second photosensitive layer **1200**. That the third width **W3** is similar to the fourth width **W4** may mean that a difference between the third width **W3** and the fourth width **W4** is less than about 10% of the third width **W3** or less than about 10% of the fourth width **W4**.

(34) The exposed color material layer **180M** may be developed. Through the developing, a pixel defining layer **180** including a first opening **1801** and a second opening **1802** may be formed above the substrate **100** as illustrated in FIG. 5.

(35) The pixel defining layer **180** may have a color material. For example, the pixel defining layer **180** may have a color pigment, for example, a pigment of a certain color such as a white color or a black color. In one embodiment, the pixel defining layer **180** may have a black color. For example, the pixel defining layer **180** may include a polyimide (PI)-based binder and a pigment in which a red color, a green color, and a blue color are mixed. As another example, the pixel defining layer **180** may include a cardo-based binder resin and a mixture of a lactam black pigment and a blue pigment. As another example, the pixel defining layer **180** may include carbon black.

(36) Since the pixel defining layer **180** is formed from the color material layer **180M**, the pixel defining layer **180** may have negative photosensitivity. Since the pixel defining layer **180** includes the color pigment, the pixel defining layer **180** may prevent reflection of external light. The pixel defining layer **180** including the color pigment may prevent reflection of external light that travels from the outside toward the light-emitting display apparatus and may improve the contrast of the light-emitting display apparatus.

(37) The substrate **100** on which the pixel defining layer **180**, the first photosensitive layer **1100**, and the second photosensitive layer **1200** are disposed may be entirely exposed. Since the first photosensitive layer **1100** and the second photosensitive layer **1200** include a positive photosensitive material, the first photosensitive layer **1100** and the second photosensitive layer **1200** are easily dissolved in a developing solution by the above-described exposure. The first photosensitive layer **1100** and the second photosensitive layer **1200** may be developed, and the first photosensitive layer **1100** and the second photosensitive layer **1200** may be removed as illustrated in FIG. 6. After the first photosensitive layer **1100** and the second photosensitive layer **1200** are removed through the developing, a curing process may be performed at a certain temperature.

(38) The edge portion of the pixel electrode **210** may be covered with the pixel defining layer **180**. The edge portion of the pixel electrode **210** may be in direct contact with the pixel defining layer **180** while overlapping the pixel defining layer **180**. A portion of the pixel electrode **210** covered with the first photosensitive layer **1100** may be exposed when the first photosensitive layer **1100** is removed. A central portion of the pixel electrode **210** may overlap the first opening **1801** of the pixel defining layer **180** and may be exposed through the first opening **1801**.

(39) An edge portion of the wiring **220** may be covered with the pixel defining layer **180**. The edge portion of the wiring **220** may overlap the pixel defining layer **180** and may be in direct contact with the pixel defining layer **180**. A portion of the wiring **220** covered with the second photosensitive layer **1200** may be exposed when the second photosensitive layer **1200** is removed. A central portion of the wiring **220** may overlap the second opening **1802** of the pixel defining layer **180** and may be exposed through the second opening **1802**.

(40) A size of the first opening **1801** may be smaller than a size of the pixel electrode **210**, and a size of the second opening **1802** may be smaller than a size of the wiring **220**. In this regard, FIG. 6 illustrates that a fifth width **W5** of the first opening **1801** is smaller than a sixth width **W6** of the pixel electrode **210**, and a seventh width **W7** of the second opening **1802** is smaller than an eighth width **W8** of the wiring **220**.

(41) Since the region of the pixel electrode **210** exposed through the first opening **1801** is covered

with the first photosensitive layer **1100** in the process described above with reference to FIGS. **1** to **6**, the region of the pixel electrode **210** is not in contact with the color material constituting the pixel defining layer **180**.

(42) As a comparative example, there is a method of forming a pixel defining layer by entirely coating a material for forming the pixel defining layer on a pixel electrode, while omitting a process of forming a first photosensitive layer and a second photosensitive layer, and then exposing and developing the coated material. A portion of the pixel defining layer **180** positioned in a region corresponding to the central portion of the pixel electrode **210** is removed so as to form the first opening, but a residue that is not removed remains on the pixel electrode **210**. This residue may cause dark spots. In order to minimize the generation of dark spots, a descum process may be performed to remove the residue. However, when the descum process is performed, the generation of dark spots at the central portion of the pixel electrode **210** is reduced, but dark spots are still generated at the edge portion of the pixel electrode **210**, for example, the edge portion of the first opening **1801**.

(43) However, according to an embodiment, as the process described above with reference to FIGS. **1** to **6** is performed, the central portion of the pixel electrode **210** is not in contact with the material constituting the pixel defining layer **180**. Therefore, it is unnecessary to perform the descum process described above, and it is possible to prevent the generation of dark spots caused by the residue of the material constituting the pixel defining layer **180**.

(44) FIGS. **7A** to **7C** are schematic cross-sectional views illustrating a part of the process of manufacturing the light-emitting display apparatus, according to an embodiment.

(45) In the process described above with reference to FIG. **4**, due to factors such as alignment between the substrate **100** and the mask **1400**, exposure time, and exposure amount, the color material layer **180M** may overlap the edge portion of the first photosensitive layer **1100** as illustrated in FIG. **7A**. Similarly, the pixel defining layer **180** may overlap the edge portion of the second photosensitive layer **1200**.

(46) As described above with reference to FIG. **5**, when the substrate **100** is entirely exposed and then developed, the first photosensitive layer **1100** and the second photosensitive layer **1200** may be removed and the pixel defining layer **180** may include a reverse-tapered inclined surface as illustrated in FIG. **7B** (see portion A and portion B of FIG. **7B**). In case that the pixel defining layer **180** includes the reverse-tapered surface, a functional layer and/or an opposite electrode to be formed in a subsequent process may be broken. However, according to an embodiment, a part of the pixel defining layer **180** may be reflowed through a curing process that is performed after the first photosensitive layer **1100** and the second photosensitive layer **1200** are removed. Therefore, as illustrated in FIG. **7C**, the portions of the pixel defining layer **180** defining the first opening **1801** and the second opening **1802** may include forward-tapered inclined surfaces while being in contact with or connected to the upper surface of the pixel electrode **210** and the upper surface of the wiring **220**.

(47) FIG. **8** is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment, which shows a pixel electrode and a wiring. The plan view of FIG. **8** may correspond to the plan view when the pixel electrode **210** and the wiring **220** formed in the process described above with reference to FIG. **2** are seen in a direction perpendicular to the substrate **100**. For convenience of description, the first photosensitive layer **1100** and the second photosensitive layer **1200** illustrated in FIG. **2** are omitted in FIG. **8**.

(48) The pixel electrode **210** is an island type and may be spaced apart from the wiring **220**. Referring to FIGS. **2** and **8**, the pixel electrode **210** may include a first portion **210A** for electrical connection with the thin-film transistor TFT, and a second portion **210B** connected to the first portion **210A**. The first portion **210A** is a portion for electrical connection with the thin-film transistor TFT disposed below the pixel electrode **210**, and the first portion **210A** may be connected to the thin-film transistor TFT through the contact hole CNT. The second portion **210B** may be

integrally connected to the first portion **210A** and may have a relatively larger area than that of the first portion **210A**.

(49) FIG. **9** is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment, which shows the pixel defining layer on the pixel electrode and the wiring. The plan view of FIG. **9** may correspond to the plan view when the cross-sectional structure described above with reference to FIGS. **6** and **7C** is seen in a direction perpendicular to the substrate.

(50) The pixel defining layer **180** may be disposed on the pixel electrode **210** and the wiring **220** and may cover the edge portion of the pixel electrode **210** and the edge portion of the wiring **220**. The first opening **1801** of the pixel defining layer **180** overlaps the pixel electrode **210**, and the second opening **1802** overlaps the wiring **220**.

(51) In the process described above with reference to FIGS. **1** to **6** and **7A** to **7C**, the pixel electrode **210** and the wiring **220** may be formed by the first photosensitive layer **1100** and the second photosensitive layer **1200**, and the pixel defining layer **180** having the first opening **1801** and the second opening **1802** may be formed by using the ashed first photosensitive layer **1100** and the ashed second photosensitive layer **1200**. Therefore, when seen in a plan view, the first opening **1801** has a size different from that of the pixel electrode **210**, but may have substantially the same shape as that of the pixel electrode **210**. Similarly, when seen in a plan view, the second opening **1802** has a size different from that of the wiring **220**, but may have substantially the same planar shape as that of the wiring **220**.

(52) The first opening **1801** may also overlap the contact hole CNT for electrical connection between the pixel electrode **210** and the thin-film transistor TFT (FIGS. **6** and **7C**). Edges of the first opening **1801** may be spaced apart from edges of the pixel electrode **210**. For example, a first edge of the first opening **1801** and a first edge of the pixel electrode **210**, which are adjacent to each other, may be spaced apart by a first distance  $d1$ , and a second edge of the first opening **1801** and a second edge of the pixel electrode **210** may be spaced apart by a second distance  $d2$ . A third edge of the first opening **1801** and a third edge of the pixel electrode **210** may be spaced apart by a third distance  $d3$ , a fourth edge of the first opening **1801** and a fourth edge of the pixel electrode **210** may be spaced apart by a fourth distance  $d4$ , and a fifth edge of the first opening **1801** and a fifth edge of the pixel electrode **210** may be spaced apart by a fifth distance  $d5$ . Since the pixel defining layer **180** is formed by using the first photosensitive layer **1100** (FIG. **2**) that is smaller than the pixel electrode **210** through the ashing as described above, the first to fifth distances  $d1$ ,  $d2$ ,  $d3$ ,  $d4$ , and  $d5$  described above may be substantially the same as each other.

(53) Similarly, one edge of the second opening **1802** may be spaced apart from one edge of the wiring **220** by a sixth distance  $d6$ , and the other edge of the second opening **1802** may be spaced apart from the other edge of the wiring **220** by a seventh distance  $d7$ . The sixth distance  $d6$  and the seventh distance  $d7$  may be substantially the same as each other.

(54) FIGS. **10A** to **10C** are schematic cross-sectional views illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment.

(55) Referring to FIG. **10A**, a spacer **191** may be formed on the pixel defining layer **180** described above with reference to FIG. **6** or **7C**. The spacer **191** may include a material different from that of the pixel defining layer **180**. In one embodiment, the spacer **191** may include a polyimide-based resin, for example, a photosensitive polyimide-based resin. The spacer **191** may overlap a material portion other than the opening of the pixel defining layer **180**.

(56) In the process of forming the spacer **191**, an insulating layer **192** may be formed to cover the second opening **1802**. The insulating layer **192** may include a same material as that of the spacer **191**. The insulating layer **192** may cover the wiring **220** that is exposed through the second opening **1802**. In an embodiment, when the intermediate layer (see FIG. **10B**) is formed on the wiring **220** exposed by not forming the insulating layer **192**, it may be impossible to provide a signal to be provided to the pixel circuit including the thin-film transistor TFT and the storage capacitor Cap

through the wiring **220**. Therefore, the insulating layer **192** may be formed to cover the wiring **220**. (57) As illustrated in FIG. **10B**, an intermediate layer **224** and an opposite electrode **230** may be formed. The intermediate layer **224** may include a first functional layer **221**, an emission layer **222**, and a second functional layer **223**, which are sequentially stacked. The emission layer **222** may overlap only the pixel electrode **210** and/or the first opening **1801**, but the first functional layer **221** and the second functional layer **223** may be formed to cover the entire substrate **100**.

(58) The emission layer **222** may include an organic material. The emission layer **222** may include a high-molecular-weight organic material or a low-molecular-weight organic material that emits light of a certain color. The emission layer **222** may be formed through a deposition process using a mask.

(59) The first functional layer **221** may be a single layer or a multi-layer. In case that the first functional layer **221** is formed of a high-molecular-weight material, the first functional layer **221** is a hole transport layer (HTL) having a single-layered structure and may be formed of poly-(3,4)-ethylene-dihydroxy thiophene (PEDOT) or polyaniline (PANI). In case that the first functional layer **221** is formed of a low-molecular-weight material, the first functional layer **221** may include a hole injection layer (HIL) and an HTL. The first functional layer **221** may be formed by, for example, thermal deposition.

(60) The second functional layer **223** may be optional. In case that the first functional layer **221** and the emission layer **222** are formed of a high-molecular-weight material, the second functional layer **223** may be formed. The second functional layer **223** may be a single layer or a multi-layer. The second functional layer **223** may include an electron transport layer (ETL) and/or an electron injection layer (EIL). The second functional layer **223** may be formed by, for example, thermal deposition.

(61) The opposite electrode **230** may be formed of a conductive material having a relatively low work function. For example, the opposite electrode **230** may include a (semi)transparent layer including silver (Ag), magnesium (Mg), aluminum (Al), nickel (Ni), chromium (Cr), lithium (Li), calcium (Ca), and/or any alloy thereof. Alternatively, the opposite electrode **230** may further include a layer such as ITO, IZO, ZnO, or In.sub.2O.sub.3 on the (semi)transparent layer including the above-mentioned material. In one embodiment, the opposite electrode **230** may include silver (Ag) and magnesium (Mg). In one embodiment, the opposite electrode **230** may be formed by thermal deposition.

(62) A stacked structure, in which the pixel electrode **210**, the intermediate layer **224**, and the opposite electrode **230** are sequentially stacked, may form a light-emitting diode, for example, an organic light-emitting diode OLED.

(63) Referring to FIG. **10C**, a thin-film encapsulation layer **300** and an optical functional layer **400** may be formed on the organic light-emitting diode OLED.

(64) The thin-film encapsulation layer **300** may include at least one inorganic encapsulation layer and at least one organic encapsulation layer. In one embodiment, the thin-film encapsulation layer **300** may include first and second inorganic encapsulation layers **310** and **330** and an organic encapsulation layer **320** therebetween.

(65) The first and second inorganic encapsulation layers **310** and **330** may each include one or more inorganic insulating materials. The inorganic insulating materials may include aluminum oxide, titanium oxide, tantalum oxide, hafnium oxide, zinc oxide, silicon oxide, silicon nitride, and/or silicon oxynitride. The first and second inorganic encapsulation layers **310** and **330** may be formed through chemical vapor deposition.

(66) The organic encapsulation layer **320** may include a polymer-based material. The polymer-based material may include an acrylic resin, an epoxy-based resin, polyimide, polyethylene, or the like. For example, the organic encapsulation layer **320** may include an acrylic resin such as polymethyl methacrylate, polyacrylic acid, or the like. The organic encapsulation layer **320** may be formed by curing a monomer or coating a polymer.

(67) The optical functional layer **400** is an element that prevents reflection of external light and may include a color filter **410** and a black matrix **420**. The light-emitting display apparatus including the optical functional layer **400** including the color filter **410** and the black matrix **420** may significantly reduce the thickness of the light-emitting display apparatus, as compared to a display apparatus including a polarizing plate. In an embodiment, the black matrix **400** may be adjacent to the color filter **410**. In another embodiment, the black matrix **400** may surround the color filter **410**.

(68) The color filter **410** may overlap the first opening **1801** of the pixel defining layer **180**, and light emitted from the organic light-emitting diode OLED passes through the thin-film encapsulation layer **300** and the color filter **410**.

(69) The black matrix **420** may overlap the wiring **220** and the second opening **1802**, and a part of the black matrix **420** may overlap a part of the first opening **1801**. For example, the black matrix **420** may overlap a part of the first opening **1801** and the contact hole CNT. When seen in a direction perpendicular to the substrate **100**, the contact hole CNT overlaps the black matrix **420** and is covered with the black matrix **420**.

(70) FIG. **11** is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment. FIG. **11** illustrates the structure of the optical functional layer **400** formed on the plan view illustrated in FIG. **9**.

(71) Referring to FIG. **11**, the color filter **410** of the optical functional layer **400** may overlap the first opening **1801**. A ninth width **W9** of the color filter **410**, which is measured in one direction, may be equal to or smaller than the fifth width **W5** of the first opening **1801**, which is measured in the one direction as described above. In one embodiment, FIG. **11** illustrates that the ninth width **W9** is smaller than the fifth width **W5**.

(72) The black matrix **420** may overlap the edge portion of the pixel electrode **210** and the contact hole CNT. In addition, the black matrix **420** may overlap the wiring **220**, the second opening **1802** overlapping the wiring **220**, and the insulating layer **192** covering the second opening **1802**.

(73) FIG. **12** is a schematic diagram of an equivalent circuit illustrating a pixel circuit connected to a light-emitting diode of the light-emitting display apparatus, for example, an organic light-emitting diode, according to an embodiment.

(74) Referring to FIG. **12**, the pixel circuit may include thin-film transistors and a storage capacitor. According to an embodiment, as illustrated in FIG. **12**, the thin-film transistors may include a driving thin-film transistor **T1**, a switching thin-film transistor **T2**, a compensation thin-film transistor **T3**, a first initialization thin-film transistor **T4**, an operation control thin-film transistor **T5**, an emission control thin-film transistor **T6**, and a second initialization thin-film transistor **T7**.

(75) A gate electrode of the driving thin-film transistor **T1** is electrically connected to an electrode of the storage capacitor Cap. One of a source electrode and a drain electrode of the driving thin-film transistor **T1** is electrically connected to a driving voltage line PL through the operation control thin-film transistor **T5**. The other of the source electrode and the drain electrode of the driving thin-film transistor **T1** is electrically connected to the pixel electrode of the organic light-emitting diode OLED through the emission control thin-film transistor **T6**. The driving thin-film transistor **T1** receives a data signal **Dm** according to a switching operation of the switching thin-film transistor **T2** and supplies a driving current **Id** to the organic light-emitting diode OLED.

(76) A gate electrode of the switching thin-film transistor **T2** is electrically connected to a first scan line SWL. One of a source electrode and a drain electrode of the switching thin-film transistor **T2** is electrically connected to a data line DL. The other of the source electrode and the drain electrode of the switching thin-film transistor **T2** is electrically connected to the driving thin-film transistor **T1** and electrically connected to the driving voltage line PL through the operation control thin-film transistor **T5**. The switching thin-film transistor **T2** is turned on according to a scan signal GW received through the first scan line SWL and performs a switching operation to transfer the data signal **Dm** received through the data line DL to the driving thin-film transistor **T1**.

- (77) A gate electrode of the compensation thin-film transistor T3 is electrically connected to the first scan line SWL. One of a source electrode and a drain electrode of the compensation thin-film transistor T3 is electrically connected to the driving thin-film transistor T1 and electrically connected to the pixel electrode of the organic light-emitting diode OLED through the emission control thin-film transistor T6. The other of the source electrode and the drain electrode of the compensation thin-film transistor T3 is electrically connected to the electrode of the storage capacitor Cap, the first initialization thin-film transistor T4, and the driving thin-film transistor T1. The compensation thin-film transistor T3 is turned on according to the scan signal GW received through the first scan line SWL and electrically connects the gate electrode of the driving thin-film transistor T1 to one of the source electrode and the drain electrode thereof (for example, the drain electrode) to diode-connect the driving thin-film transistor T1.
- (78) A gate electrode of the first initialization thin-film transistor T4 is electrically connected to a second scan line SIL. One of a source electrode and a drain electrode of the first initialization thin-film transistor T4 is electrically connected to the second initialization thin-film transistor T7 and an initialization voltage line VL. The other of the source electrode and the drain electrode of the first initialization thin-film transistor T4 is electrically connected to the electrode of the storage capacitor Cap, the compensation thin-film transistor T3, and the driving thin-film transistor T1. The first initialization thin-film transistor T4 is turned on according to a previous scan signal GI received through the second scan line SIL and performs an initialization operation to transfer an initialization voltage Vint to the gate electrode of the driving thin-film transistor T1 to initialize the voltage of the gate electrode of the driving thin-film transistor T1.
- (79) A gate electrode of the operation control thin-film transistor T5 is electrically connected to an emission control line EL. One of a source electrode and a drain electrode of the operation control thin-film transistor T5 is electrically connected to the driving voltage line PL. The other of the source electrode and the drain electrode of the operation control thin-film transistor T5 is electrically connected to the driving thin-film transistor T1 and the switching thin-film transistor T2.
- (80) A gate electrode of the emission control thin-film transistor T6 is electrically connected to the emission control line EL. One of a source electrode and a drain electrode of the emission control thin-film transistor T6 is electrically connected to the driving thin-film transistor T1 and a compensation source electrode of the compensation thin-film transistor T3. The other of the source electrode and the drain electrode of the emission control thin-film transistor T6 is electrically connected to the second initialization thin-film transistor T7 and the pixel electrode of the organic light-emitting diode OLED.
- (81) The operation control thin-film transistor T5 and the emission control thin-film transistor T6 are simultaneously turned on according to an emission control signal En received through the emission control line EL and transfers a driving voltage ELVDD to the organic light-emitting diode OLED so that the driving current Id flows through the organic light-emitting diode OLED.
- (82) A gate electrode of the second initialization thin-film transistor T7 is electrically connected to the second scan line SIL. One of a source electrode and a drain electrode of the second initialization thin-film transistor T7 is electrically connected to the emission control thin-film transistor T6 and the pixel electrode of the organic light-emitting diode OLED. The other of the source electrode and the drain electrode of the second initialization thin-film transistor T7 is electrically connected to the first initialization thin-film transistor T4 and the initialization voltage line VL. The second initialization thin-film transistor T7 is turned on according to a previous scan signal GI received through the second scan line SIL and initializes the pixel electrode of the organic light-emitting diode OLED.
- (83) An embodiment where the first initialization thin-film transistor T4 and the second initialization thin-film transistor T7 are electrically connected to the second scan line SIL is illustrated in FIG. 12, but embodiments are not limited thereto. In another embodiment, the first

initialization thin-film transistor T4 may be electrically connected to the second scan line SIL and driven according to the previous scan signal GI, the second initialization thin-film transistor T7 may be electrically connected to the first scan line or the second scan line of the pixel arranged (or disposed) in a previous row or a next row of the pixel P.

(84) One electrode of the storage capacitor Cap is electrically connected to the driving voltage line PL, and the opposite electrode of the organic light-emitting diode OLED is electrically connected to a common voltage ELVSS. Therefore, the organic light-emitting diode OLED may receive the driving current Id from the driving thin-film transistor T1 and emit light to display an image.

(85) An embodiment where the pixel circuit PC includes seven thin-film transistors and one storage capacitor has been described with reference to FIG. 12, but embodiments are not limited thereto. The number of thin-film transistors and the number of storage capacitors may be variously changed according to the design of the pixel circuit PC.

(86) The wiring 220 described above with reference to FIG. 11 may be one of the lines described above with reference to FIG. 12, for example, the driving voltage line PL, the initialization voltage line VL, and the data line DL. For example, the wiring 220 may correspond to the initialization voltage line VL configured to provide the initialization voltage Vint.

(87) In another embodiment, the wiring 220 may correspond to a line configured to provide signals or voltages different from those of the lines (for example, the driving voltage line, the initialization voltage line, the data line, and the like) connected to the pixel circuit. The structure of the cross section and the plane on the wiring 220 may be different from the structure described above with reference to FIGS. 10C and 11, and the structure will be described with reference to FIGS. 13A to 13C and 14.

(88) FIGS. 13A to 13E are schematic cross-sectional views illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment.

(89) Referring to FIG. 13A, a spacer 191 may be formed on the pixel defining layer 180 described above with reference to FIG. 6 or 7C. The spacer 191 may include a material different from that of the pixel defining layer 180. In one embodiment, the spacer 191 may include a polyimide-based resin, for example, a photosensitive polyimide-based resin. The spacer 191 may overlap a material portion other than the opening of the pixel defining layer 180, for example, the first opening 1801 and the second opening 1802.

(90) As illustrated in FIG. 13B, an intermediate layer 224 may be formed. The intermediate layer 224 may include a first functional layer 221, an emission layer 222, and a second functional layer 223, which are sequentially stacked. The emission layer 222 may overlap the pixel electrode 210, but the first functional layer 221 and the second functional layer 223 may be formed to cover the entire substrate 100. Materials of the emission layer 222, the first functional layer 221, and the second functional layer 223 are the same as those described above.

(91) A part of the intermediate layer 224, for example, the first functional layer 221 and/or the second functional layer 223, may be positioned on the wiring 220 through the second opening 1802. The first functional layer 221 may be in direct contact with the wiring 220 through the second opening 1802.

(92) As illustrated in FIG. 13C, a portion of the first functional layer 221 on the wiring 220 and a portion of the second functional layer 223 on the wiring 220 may be removed to form a first hole 221h of the first functional layer 221 and a second hole 223h of the functional layer 223. The first hole 221h and the second hole 223h may overlap the second opening 1802, and the wiring 220 may be exposed through the first hole 221h and the second hole 223h. The first hole 221h and the second hole 223h may be formed by irradiating a laser beam.

(93) As illustrated in FIG. 13D, an opposite electrode 230 is formed. A material of the opposite electrode 230 is the same as that described above.

(94) A stacked structure, in which the pixel electrode 210, the intermediate layer 224, and the opposite electrode 230 are sequentially stacked, may form a light-emitting diode, for example, an

organic light-emitting diode OLED. A part of the opposite electrode **230** may be electrically connected to the wiring **220**. A part of the opposite electrode **230** may be in direct contact with the wiring **220** exposed through the first hole **221h** and the second hole **223h**. The wiring **220** may be a bus line capable of providing the common voltage ELVSS (FIG. **12**). When the light-emitting display apparatus has a relatively large size, bus lines may be disposed adjacent to pixel electrodes in a display area so as to prevent voltage drop of the opposite electrode. The display area refers to an image plane or a screen area on which the light-emitting display apparatus is capable of providing an image. The wiring **220** illustrated in FIG. **13D** may correspond to the bus line described above and may be configured to provide the common voltage ELVSS to the opposite electrode **230**.

(95) As illustrated in FIG. **13E**, a thin-film encapsulation layer **300** and an optical functional layer **400** may be formed on the organic light-emitting diode OLED. The detailed structures of the thin-film encapsulation layer **300** and the optical functional layer **400** are the same as those described above with reference to FIG. **10C**.

(96) The color filter **410** may overlap the first opening **1801** of the pixel defining layer **180**, and light emitted from the organic light-emitting diode OLED passes through the thin-film encapsulation layer **300** and the color filter **410**.

(97) The black matrix **420** may overlap the wiring **220** and the second opening **1802**, and a part of the black matrix **420** may overlap a part of the first opening **1801**. The black matrix **420** may overlap a part of the first opening **1801** and the contact hole CNT. When seen in a direction perpendicular to the substrate **100**, the contact hole CNT overlaps the black matrix **420** and is covered with the black matrix **420**.

(98) FIG. **14** is a plan view illustrating the process of manufacturing the light-emitting display apparatus, according to an embodiment. FIG. **14** may correspond to a planar structure when the structure illustrated in FIG. **13E** is seen in a direction perpendicular to the substrate **100**.

(99) Referring to FIG. **14**, the color filter **410** of the optical functional layer **400** may overlap the first opening **1801**. The ninth width **W9** of the color filter **410**, which is measured in the same direction, may be equal to or smaller than the fifth width **W5** of the first opening **1801**. In one embodiment, FIG. **14** illustrates that the ninth width **W9** is smaller than the fifth width **W5**.

(100) The black matrix **420** may overlap the edge portion of the pixel electrode **210** and the contact hole CNT. The black matrix **420** may overlap the wiring **220**, the second opening **1802** overlapping the wiring **220**, and the first hole **221h** and the second hole **223h** overlapping the second opening **1802**.

(101) The embodiments may provide the light-emitting display apparatus capable of preventing generation of dark spots without an increase of masks and preventing deterioration of contrast.

(102) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While the embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

## Claims

1. A light-emitting display apparatus, comprising: a substrate; a thin-film transistor on the substrate; a pixel electrode electrically connected to the thin-film transistor; a first insulating layer disposed between the thin-film transistor and the pixel electrode; a wiring on the first insulating layer that is coplanar with, and spaced apart from, the pixel electrode; a pixel defining layer covering an edge portion of the pixel electrode and an edge portion of the wiring, the pixel defining layer comprising a first opening overlapping the pixel electrode and a second opening overlapping the wiring; an



intermediate layer comprising: a first functional layer on the pixel electrode and overlapping the first opening, the pixel defining layer, and the second opening; an emission layer having first and second ends, the emission layer directly on the first functional layer and overlapping the first opening; and a second functional layer on the emission layer and directly on the first functional layer beyond the first and second ends of the emission layer; and an opposite electrode on the intermediate layer, wherein, when seen in a direction perpendicular to the substrate, (i) the wiring comprises a pair of first side edges that are parallel to each other and extend in a direction perpendicular to a width direction of the wiring, the pixel defining layer comprises a pair of second side edges that are spaced apart from each other to define the first opening, the pixel defining layer further comprises a pair of third side edges that are spaced apart from each other to define the second opening entirely exposing the wiring between the pair of third side edges, and the first pair of side edges of the wiring and the pair of third side edges of the pixel defining layer are parallel to each other, and (ii) the second opening defined by the pair of third side edges of the pixel defining layer extends entirely between the pair of third side edges in the direction perpendicular to the width direction of the wiring and a part of the second opening is bent with an obtuse angle; wherein the light-emitting display apparatus further comprises a second insulating layer having an upper surface coplanar with an upper surface of the pixel defining layer in a thickness direction, the second insulating layer overlapping the second opening.

2. The light-emitting display apparatus of claim 1, wherein the first insulating layer comprises a contact hole for electrical connection between the thin-film transistor and the pixel electrode, and the first opening overlaps the contact hole.

3. The light-emitting display apparatus of claim 2, further comprising: an optical functional layer comprising a color filter overlapping the first opening; and a black matrix adjacent to the color filter and overlapping a part of the first opening.

4. The light-emitting display apparatus of claim 3, wherein the black matrix overlaps the contact hole.

5. The light-emitting display apparatus of claim 3, wherein the second opening is overlapped by the black matrix, and the part of the second opening bent with the obtuse angle is overlapped by the black matrix.

6. The light-emitting display apparatus of claim 1, wherein the pixel defining layer further comprises a color pigment or carbon black.

7. The light-emitting display apparatus of claim 1, wherein a shape of the first opening is substantially same as a shape of the pixel electrode when seen in the direction perpendicular to the substrate.

8. The light-emitting display apparatus of claim 1, wherein a first distance from a first edge of the pixel electrode to a first edge of the first opening is substantially same as a second distance from a second edge of the pixel electrode to a second edge of the first opening.

9. The light-emitting display apparatus of claim 1, wherein the wiring and the pixel electrode include a same material.

10. The light-emitting display apparatus of claim 1, wherein a shape of the second opening is substantially same as a shape of the wiring when seen in the direction perpendicular to the substrate.

11. The light-emitting display apparatus of claim 1, the second insulating layer directly contacting the wiring in the second opening and directly contacting the pair of side edges of the pixel defining layer defining the second opening.

12. The light-emitting display apparatus of claim 11, wherein the second insulating layer is disposed between the wiring and the intermediate layer.

13. The light-emitting display apparatus of claim 11, further comprising: a spacer disposed on the pixel defining layer, wherein the second insulating layer and the spacer comprise a same material.

14. The light-emitting display apparatus of claim 13, wherein the pixel defining layer and the

spacer comprise different materials.

15. The light-emitting display apparatus of claim 11, further comprising: a driving thin-film transistor electrically connected to the thin-film transistor; an initialization thin-film transistor electrically connected to a gate electrode of the driving thin-film transistor or the pixel electrode; and an initialization voltage line electrically connected to the initialization thin-film transistor, wherein the wiring comprises the initialization voltage line.

16. The light-emitting display apparatus of claim 1, wherein the first functional layer and the second functional layer each comprise a hole overlapping the second opening.

17. The light-emitting display apparatus of claim 16, wherein the opposite electrode is in electrical contact with the wiring through the hole.

18. The light-emitting display apparatus of claim 1, wherein the first and second ends of the emission layer do not extend beyond the pixel electrode.

19. The light-emitting display apparatus of claim 1, further comprising: a second insulating layer disposed in the second opening and on the wiring, wherein no emission layer covers the wiring, and the first functional layer covers the wiring.

20. The light-emitting display apparatus of claim 1, wherein the first functional layer extends to overlap the first opening, the pixel defining layer, and the second opening in its entirety.

21. The light-emitting display apparatus of claim 1, wherein the first functional layer directly contacts the pair of third side edges and the wiring, and the opposite electrode directly contacts the wiring between the third side edges.

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